

# From Unequal Growth to Skirmish

## An Integrated Economic, Geopolitical, Militaric, and Statistical Theory

Lord Soumadeep Ghosh and Leader of Opposition Rahul Gandhi  
with ChatGPT (OpenAI)

Kolkata, India

### Abstract

This paper develops an integrated framework connecting regional economic growth, geopolitical influence, militaric capability, and the emergence and dynamics of interstate skirmishes. The framework synthesizes three preceding research programmes: the Ghosh Factor for modelling unequal economic growth, a geometric theory of geopolitical influence based on military, economic, and demographic observables, and a model of skirmish based on the divergence between geopolitical and militaric scores. The resulting architecture treats economic development as an upstream state variable which changes the resource base of geopolitical projection and militaric sustainability.

The central object is the skirmish gap between geopolitical influence and militaric capacity. The paper extends this gap into a statistical conflict-risk framework involving panel econometrics, survival analysis, graph learning, Bayesian inference, and machine learning. Economic and financial variables are interpreted as state-capacity variables, while geography and diplomatic relations provide the network structure through which geopolitical pressure propagates. Psychological and philosophical considerations are incorporated only insofar as perception, bounded rationality, legitimacy, and uncertainty affect observable strategic behaviour.

The framework is theoretical and provides an empirical research programme. It does not establish that unequal growth causes war, nor does it establish that a particular state or government is responsible for a particular conflict. Those propositions require independent longitudinal data and out-of-sample testing.

The paper ends with “The End”

## 1 Introduction

Economic development, military capability, geopolitical influence, and armed conflict are frequently studied as separate objects. Economic growth is generally treated as a development phenomenon, military capability as a strategic phenomenon, and conflict as an international-relations phenomenon. The present paper proposes that these objects can instead be placed inside a single dynamic system.

The starting point is the observation that economic resources constrain both the ability of a state to project geopolitical influence and its ability to sustain military capability. A state may therefore experience a divergence between the geopolitical role that it projects or is perceived to possess and the militaric capacity that its economic and demographic structure can sustain.

Let  $W_i(t)$  denote wealth,  $P_i(t)$  population,  $M_i(t)$  military size, and  $m_i(t)$  the wealth absorbed by the military of state  $i$ . The integrated system is represented schematically by

$$\text{Economic Growth} \longrightarrow \text{Economic Capacity} \longrightarrow \{Z_i, Y_i\} \longrightarrow \Delta_i \longrightarrow S_{ij} \longrightarrow H_{ij}. \quad (1)$$

Here  $Z_i$  is geopolitical influence,  $Y_i$  is militaric capability,  $\Delta_i$  is geopolitical–militaric misalignment,  $S_{ij}$  is observable skirmish intensity, and  $H_{ij}$  denotes a future conflict hazard.

The contribution is therefore not a claim that any single variable determines war. Rather, it is a proposed architecture through which several measurable state variables can interact.

## 2 Economic Foundations

### 2.1 Unequal regional growth

Let  $g_i$  denote the GDP growth rate of region  $i$ , and let  $\bar{g}$  denote the appropriate group benchmark. The regional growth deviation is

$$D_i = g_i - \bar{g}. \quad (2)$$

The Ghosh Factor is a conditional logarithmic transformation:

$$G_X = \begin{cases} \log(X), & D_i < 0, \\ \log(1 + X), & D_i \geq 0. \end{cases} \quad (3)$$

The preceding empirical application used official GDP-growth data for all 28 Indian states and 8 Union Territories for 2023–24, with constant-price GSDP data and regional deviations from group averages.

A parsimonious econometric specification is

$$D_i = \alpha + \gamma G_{X_i} + \varepsilon_i. \quad (4)$$

The national growth rate is constant across the cross-sectional observations and therefore cannot independently identify a coefficient in the regression.

The important methodological qualification is that this specification is cross-sectional. A high coefficient of determination does not establish a causal mechanism, and it does not establish temporal forecasting power.

### 2.2 From cross-sectional growth to dynamic capability

For a longitudinal application, let

$$W_i(t+1) = F(W_i(t), g_i(t), D_i(t), G_i(t), X_i(t)) + \varepsilon_i(t). \quad (5)$$

The function  $F$  need not initially be specified parametrically. It may be estimated through panel methods, state-space models, Bayesian inference, or machine learning.

The principal conceptual transition is therefore

$$g_i(t) \longrightarrow W_i(t) \longrightarrow \{Z_i(t), Y_i(t)\}. \quad (6)$$

Economic growth is consequently treated as an upstream determinant of the resource envelope within which geopolitical and militaric behaviour occurs.

## 3 Geopolitical Influence

### 3.1 The geopolitical circle

Let  $\theta_i$  denote the bearing from a reference state to neighbouring state  $i$ . The unweighted geographic direction is the circular mean

$$\theta_{\text{geo}} = \text{atan2} \left( \frac{1}{N} \sum_{i=1}^N \sin \theta_i, \frac{1}{N} \sum_{i=1}^N \cos \theta_i \right). \quad (7)$$

The geometric representation can be visualised as follows.

### 3.2 Composite geopolitical score

For each neighbouring state, let  $M_i$ ,  $W_i$ , and  $P_i$  denote military size, economic output, and population. The composite geopolitical score is

$$Z_i = \alpha \log_{10} M_i + \beta \log_{10} W_i + \gamma \log_{10} P_i, \quad (8)$$

where the parameters represent strategic weights.

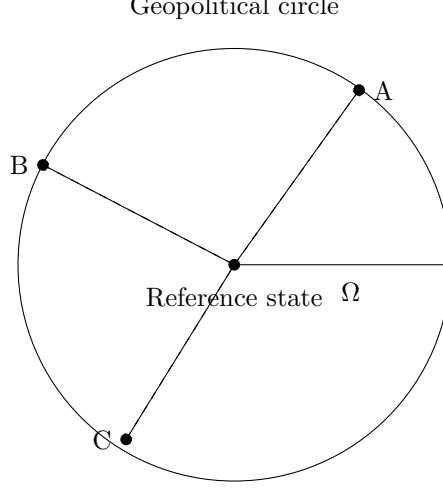


Figure 1: A schematic geopolitical circle. Directional influence is represented by bearings and subsequently weighted by geopolitical capability.

The use of logarithms imposes diminishing marginal contribution of very large absolute quantities. This is appropriate when the objective is to construct a relative capability index rather than simply sum physical resources.

The weighted direction of geopolitical influence can then be written as

$$\Theta = \text{atan2} \left( \sum_{i=1}^N Z_i \sin \theta_i, \sum_{i=1}^N Z_i \cos \theta_i \right). \quad (9)$$

Thus geography supplies the direction while capability supplies the weight.

## 4 Militaric Capability

The militaric score is specified as

$$Y_i = \alpha_M \log M_i + \beta_M \log(W_i - m_i) + \gamma_M \log(P_i - M_i). \quad (10)$$

The three components represent military prowess, economically available wealth after military absorption, and the civilian demographic base.

The requirement

$$W_i > m_i \quad (11)$$

is economically necessary for the logarithm to be defined, while

$$P_i > M_i \quad (12)$$

is required for the demographic term to remain defined.

The distinction between  $Z_i$  and  $Y_i$  is fundamental. The former is constructed as a measure of externally relevant geopolitical capability, whereas the latter incorporates the internal resource requirements of military capability.

## 5 The Geopolitical–Militaric Gap

Define the state-level misalignment as

$$\Delta_i(t) = |Z_i(t) - Y_i(t)|. \quad (13)$$

The interpretation is straightforward:

$$\Delta_i(t) \approx 0 \implies \text{geopolitical projection and militaric capacity are aligned,} \quad (14)$$

$$\Delta_i(t) \gg 0 \implies \text{greater geopolitical–militaric misalignment.} \quad (15)$$

This is not a claim that large  $\Delta_i$  mechanically causes conflict. It is a latent pressure variable whose empirical relationship with observed conflict must be estimated.

## 5.1 A simple proposition

**Proposition 1.** Suppose  $Z_i(t)$  and  $Y_i(t)$  are continuous and finite. Then  $\Delta_i(t)$  is non-negative and continuous.

**Proof.** The absolute-value function is continuous on the real line. Since the difference  $Z_i(t) - Y_i(t)$  is continuous, its absolute value is continuous. Moreover, the absolute value is non-negative. Hence  $\Delta_i(t) \geq 0$  and is continuous.  $\square$

This elementary result matters empirically because it permits the gap to enter continuous-time or discrete-time statistical models without introducing artificial discontinuities.

## 6 The Skirmish Model

For states  $A$  and  $B$ , define

$$S_A(t) = o_A |Z_A - Y_A| e^{-At}, \quad (16)$$

$$S_B(t) = o_B |Z_B - Y_B| e^{-At}. \quad (17)$$

Here  $o_A$  and  $o_B$  are coefficients of operational largeness and  $A$  is the attenuation rate. The ratio satisfies

$$\frac{S_A(t)}{S_B(t)} = \frac{o_A |Z_A - Y_A|}{o_B |Z_B - Y_B|}. \quad (18)$$

The common attenuation factor cancels from the ratio.

This provides a useful empirical separation:

$$\text{Observed intensity} = \text{structural misalignment} \times \text{operational conversion} \times \text{temporal attenuation.} \quad (19)$$

Consequently, two states can have similar geopolitical–militaric gaps but produce different observed skirmish intensities because their operational largeness differs.

## 7 Econometrics of Conflict

The integrated model naturally generates several empirical specifications.

### 7.1 Panel model

Let  $S_{ijt}$  denote an observed measure of skirmish intensity between states  $i$  and  $j$ . A panel specification is

$$S_{ijt} = \mu_{ij} + \tau_t + \beta_1 \Delta_{it} + \beta_2 \Delta_{jt} + \beta_3 |\Theta_i - \Theta_j| + \beta_4 D_{it} + \beta_5 D_{jt} + \varepsilon_{ijt}. \quad (20)$$

Here  $\mu_{ij}$  captures time-invariant dyadic heterogeneity and  $\tau_t$  captures common temporal shocks.

### 7.2 Count model

If  $S_{ijt}$  is a count of incidents, a Poisson or negative-binomial model may be appropriate:

$$E[S_{ijt} | X_{ijt}] = \exp(X'_{ijt} \beta). \quad (21)$$

The negative-binomial specification should be preferred when the data exhibit overdispersion relative to the Poisson restriction.

### 7.3 Binary escalation model

Define

$$E_{ijt+h} = \begin{cases} 1, & \text{if escalation occurs within horizon } h, \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

Then

$$\Pr(E_{ijt+h} = 1) = \frac{1}{1 + \exp(-X'_{ijt}\beta)}. \quad (23)$$

The vector  $X_{ijt}$  may include

$$X_{ijt} = (\Delta_i, \Delta_j, |\Theta_i - \Theta_j|, D_i, D_j, S_{ijt}, G_{ijt}, R_{ijt})', \quad (24)$$

where  $G_{ijt}$  represents geopolitical-network variables and  $R_{ijt}$  represents relevant economic or financial controls.

### 7.4 Survival model

For the time until escalation, define a hazard function

$$\lambda_{ij}(t) = \lambda_0(t) \exp(X_{ij}(t)'\beta). \quad (25)$$

This formulation directly addresses the question of whether a sequence of skirmishes changes the conditional hazard of subsequent escalation.

## 8 Causal Identification and the Meaning of History

A central methodological problem is that conflict observations are not randomly assigned. Military spending, economic growth, diplomatic relations, and skirmish intensity can influence one another.

A useful causal graph is

$$\text{Economic Capacity} \longrightarrow \text{Military Capacity} \longrightarrow \text{Skirmish}. \quad (26)$$

But geopolitical perception may also affect both military expenditure and skirmish probability:

$$\text{Perception} \longrightarrow \begin{cases} \text{Military Capacity} \\ \text{Skirmish} \end{cases}. \quad (27)$$

Consequently, simple correlation between  $Y$ ,  $Z$ , and skirmish counts cannot establish causality.

The historical record therefore has two distinct functions. First, it supplies observations from which statistical relationships can be estimated. Second, it contains institutional and diplomatic information required to interpret those relationships.

The past itself cannot be altered as an event. Its information content, interpretation, and consequences for future decision-making can, however, change as new evidence becomes available.

## 9 Diplomacy and International Relations

The model is compatible with classical bargaining interpretations of international relations.

A state does not act solely on material capability. It also acts on beliefs about the intentions and capabilities of other states. Hence the effective state variables can be written as

$$\tilde{Z}_{ij} = Z_j + \eta_{ij}, \quad (28)$$

where  $\eta_{ij}$  represents perception or measurement error.

Similarly,

$$\tilde{Y}_{ij} = Y_j + \nu_{ij}. \quad (29)$$

The perceived skirmish gap becomes

$$\tilde{\Delta}_{ij} = \left| \tilde{Z}_{ij} - \tilde{Y}_{ij} \right|. \quad (30)$$

Diplomatic communication can therefore have an economically meaningful role even when it does not alter physical military capacity: it may alter the information structure through which capability is perceived.

This is consistent with bargaining models of conflict in which private information, commitment problems, and misperception can influence the probability of escalation.

## 10 Psychology, Behaviour, and Decision-Making

The framework requires no assumption that states possess human psychology. Nevertheless, decision-makers inside states are human agents operating under uncertainty.

Three behavioural mechanisms are particularly relevant.

First, bounded rationality may cause decision-makers to use imperfect approximations of  $Z$  and  $Y$ .

Second, asymmetric information may generate

$$E_i[Z_j] \neq Z_j. \quad (31)$$

Third, strategic interaction may produce feedback:

$$\text{Action}_i \longrightarrow \text{Belief}_j \longrightarrow \text{Action}_j \longrightarrow \text{Belief}_i. \quad (32)$$

These mechanisms can produce conflict even when the underlying physical capabilities of the states have not changed substantially.

No psychiatric diagnosis is required or justified by this model. The relevant variables are decision-making mechanisms, information, incentives, and observable behaviour.

## 11 Welfare Economics and Warfare Economics

A government maximizes neither military capability nor GDP in isolation. Civilian welfare and security compete for finite resources.

Let national welfare be

$$\mathcal{W} = U(C, H, S) - C_M(M) - C_C(S), \quad (33)$$

where  $C$  denotes civilian consumption,  $H$  human development,  $S$  security outcomes,  $C_M$  the resource cost of military capability, and  $C_C$  the cost of conflict.

An increase in military expenditure may increase  $Y$  while simultaneously reducing the civilian resource term  $W - m$ .

Therefore the derivative

$$\frac{\partial Y}{\partial m} = -\frac{\beta_M}{W - m} \quad (34)$$

is negative whenever  $\beta_M > 0$ .

This captures an important warfare-economic trade-off: military absorption can increase military inputs while reducing the economic resources available to support the civilian population.

The optimal policy problem is consequently not

$$\max Y, \quad (35)$$

but rather

$$\max \mathcal{W}(C, H, S, Y, Z). \quad (36)$$

## 12 Finance and Political Risk

Conflict expectations can affect financial markets before physical conflict occurs.

Let  $R_{ij,t}$  denote a risk premium associated with geopolitical exposure. A reduced-form relationship is

$$R_{ij,t} = a + b\Delta_{ij,t} + cS_{ij,t} + dX_{ij,t} + u_{ij,t}. \quad (37)$$

Potential market variables include sovereign spreads, exchange rates, insurance premia, commodity transportation costs, and defense-sector valuations.

The important distinction is between prediction and causation. A financial market may price a conflict risk without causing that risk.

## 13 Graph Representation

The geopolitical system is naturally represented as a graph

$$G_t = (V_t, E_t), \quad (38)$$

where states are nodes and geopolitical relationships are edges.

Each node may contain

$$x_i(t) = [W_i, P_i, M_i, m_i, g_i, Z_i, Y_i, \Delta_i]'. \quad (39)$$

Each edge may contain

$$e_{ij}(t) = [d_{ij}, \theta_{ij}, \Delta_{ij}, S_{ij}, T_{ij}, A_{ij}]', \quad (40)$$

where  $d_{ij}$  is geographic distance and  $T_{ij}$  may represent trade or diplomatic connectivity.

A graph neural network can then approximate

$$h_i^{(l+1)} = \phi \left( W_0 h_i^{(l)} + \sum_{j \in \mathcal{N}(i)} W_1 h_j^{(l)} + W_2 e_{ij} \right). \quad (41)$$

The objective is not to replace economic theory with machine learning. It is to permit estimation of nonlinear interactions and network effects that are difficult to specify parametrically.

## 14 Bayesian Formulation

Let  $\theta$  denote the parameters of the complete model and  $D$  the historical dataset. Bayesian inference gives

$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta). \quad (42)$$

The posterior predictive distribution is

$$p(S_{t+h} \mid D) = \int p(S_{t+h} \mid \theta, D)p(\theta \mid D) d\theta. \quad (43)$$

This is particularly useful where the quantities  $o_i$ ,  $A$ , perception errors, or geopolitical weights are only partially observed.

## 15 Machine Learning and Empirical Learning

The empirical programme can be implemented in stages.

1. Construct the economic-growth panel.
2. Construct  $Z_i(t)$  and  $Y_i(t)$ .
3. Compute  $\Delta_i(t)$ .

4. Construct a geographically and diplomatically structured state graph.
5. Merge dated conflict and skirmish observations.
6. Estimate parametric benchmark models.
7. Estimate nonlinear machine-learning models.
8. Conduct temporal out-of-sample validation.
9. Compare predictive performance against simple baselines.
10. Conduct robustness, calibration, and sensitivity analysis.

A generic algorithm is:

INPUT:

- economic panel
- demographic panel
- military panel
- geographic network
- diplomatic observations
- conflict-event observations

FOR every state  $i$  and period  $t$ :

- calculate growth deviation  $D[i,t]$
- calculate Ghosh factor  $G[i,t]$
- calculate geopolitical score  $Z[i,t]$
- calculate militaric score  $Y[i,t]$
- calculate gap  $\Delta[i,t]$

FOR every dyad  $(i,j)$  and period  $t$ :

- calculate angular imbalance
- construct dyadic covariates
- attach observed skirmish outcomes
- attach subsequent escalation outcomes

SPLIT DATA TEMPORALLY:

- training period
- validation period
- test period

ESTIMATE:

- panel benchmark
- count model
- logistic model
- survival model
- Bayesian model
- graph-learning model

EVALUATE:

- log loss
- Brier score
- calibration
- ROC-AUC
- precision-recall
- out-of-sample likelihood

RETURN:

- parameter estimates
- uncertainty intervals

calibrated risk estimates  
model comparison

The temporal split is essential. Randomly mixing future observations into the training sample would create look-ahead bias.

## 16 Physics and Dynamical Interpretation

The attenuation equation

$$S(t) = S(0)e^{-At} \quad (44)$$

has the mathematical form of a first-order dissipative dynamical system. Differentiating gives

$$\frac{dS}{dt} = -AS. \quad (45)$$

Thus  $A > 0$  produces decay in the absence of renewed pressure. A more general conflict-pressure system could therefore be written as

$$\frac{dS}{dt} = \kappa\Delta(t) - AS(t). \quad (46)$$

The steady state under constant  $\Delta$  is

$$S^* = \frac{\kappa\Delta}{A}. \quad (47)$$

This interpretation is useful because it separates two mechanisms:

$$\text{new geopolitical--militaric pressure} \quad \text{versus} \quad \text{natural attenuation}. \quad (48)$$

A diplomatic intervention may therefore operate by reducing  $\Delta$ , increasing  $A$ , or both.

## 17 Information, Chemistry, and Biology as Analogical Domains

The integrated framework is primarily economic, statistical, and political. Analogies from other disciplines are therefore used cautiously.

In information theory, uncertainty can be represented by entropy:

$$H(X) = - \sum_x p(x) \log p(x). \quad (49)$$

Improved intelligence or diplomacy can be interpreted as reducing uncertainty about relevant state variables.

In biological systems, resource allocation creates trade-offs between competing functions. The analogy is useful for thinking about civilian and military resource allocation, but biological evolutionary claims should not be confused with causal mechanisms in international relations.

Similarly, chemical reaction systems provide a useful mathematical analogy for threshold processes and competing rates, but interstate conflict is not a chemical reaction and should not be modelled as one without empirical justification.

These analogies are therefore conceptual rather than evidentiary.

## 18 Philosophy, Ethics, and Religion

The framework raises an epistemological distinction between history and knowledge of history.

A past event is fixed as an event. The information available about that event is not necessarily fixed. Hence

$$\text{historical event} \neq \text{historical record} \neq \text{historical interpretation}. \quad (50)$$

This distinction is important for empirical political economy. New evidence can change the statistical representation and interpretation of historical events without changing the events themselves.

The ethical objective of the framework is correspondingly not prediction for its own sake. The relevant normative objective is the reduction of avoidable human suffering subject to legitimate political authority, civilian welfare, and the preservation of human agency.

Religious traditions frequently provide distinct conceptions of duty, restraint, sacrifice, justice, and legitimate authority. Such concepts may inform political philosophy, but they should not be inserted as empirical coefficients into the statistical model without operational definitions.

## 19 Main Integrated Proposition

The central proposition of the paper can now be stated.

**Proposition 2.** Suppose that economic capacity determines, directly or indirectly, the evolution of geopolitical and militaric scores, and that skirmish intensity is increasing in their divergence. Then economic development becomes an upstream state variable in a dynamic model of conflict pressure.

Formally, suppose

$$Z_i(t) = f_Z(W_i(t), M_i(t), P_i(t)), \quad (51)$$

$$Y_i(t) = f_Y(W_i(t), M_i(t), P_i(t), m_i(t)), \quad (52)$$

$$\Delta_i(t) = |Z_i(t) - Y_i(t)|, \quad (53)$$

$$S_i(t) = f_S(\Delta_i(t), o_i, A_i, t). \quad (54)$$

If economic development changes  $W_i(t)$ ,  $M_i(t)$ , or  $P_i(t)$ , then economic development changes the arguments of  $f_Z$ ,  $f_Y$ , and therefore potentially  $S_i(t)$ .

**Proof.** By composition of functions, changes in any argument of  $f_Z$  or  $f_Y$  can change  $Z_i$  or  $Y_i$ . Since  $\Delta_i$  is a function of these two quantities, changes in either score can change  $\Delta_i$ . Since  $S_i$  is a function of  $\Delta_i$ , the resulting change can propagate to observable skirmish intensity. The proposition establishes a transmission mechanism, not a sign for every empirical coefficient.  $\square$

## 20 Empirical Identification Strategy

The complete empirical programme should distinguish four questions.

1. Does unequal economic growth predict changes in state capability?
2. Do changes in capability predict changes in  $Z$  and  $Y$ ?
3. Does the gap  $\Delta$  predict observed skirmish intensity?
4. Does the historical trajectory of these variables improve out-of-sample prediction of escalation?

The fourth question is the strongest test.

A useful hierarchy is therefore:

Table 1: Empirical hierarchy

| Level | Question       | Preferred evidence                 |
|-------|----------------|------------------------------------|
| I     | Unequal growth | Cross-sectional/panel econometrics |
| II    | Capability     | State-level longitudinal data      |
| III   | Misalignment   | $Z, Y, \Delta$ estimation          |
| IV    | Skirmish       | Incident-level panel/count models  |
| V     | Escalation     | Survival/hazard models             |
| VI    | Prediction     | Temporal out-of-sample testing     |

No higher level should be treated as empirically established merely because a lower level has been estimated successfully.

## 21 Data Requirements

A serious implementation requires a longitudinal database containing, where legally and practically available,

Table 2: Core empirical variables

| Domain     | Variables                                 | Frequency        |
|------------|-------------------------------------------|------------------|
| Economy    | GDP, growth, wealth                       | Annual/quarterly |
| Demography | Population, demographic structure         | Annual           |
| Military   | Personnel, expenditure, equipment proxies | Annual/quarterly |
| Geography  | Bearings, distance, borders               | Static/dynamic   |
| Diplomacy  | Agreements, disputes, negotiations        | Event-level      |
| Skirmishes | Date, location, participants, intensity   | Event-level      |
| Casualties | Military and civilian losses              | Event-level      |
| Finance    | Spreads, FX, insurance, commodities       | Daily/monthly    |
| Conflict   | Escalation/de-escalation outcomes         | Event-level      |

The distinction between public and restricted information is also important. The model should never assume that an unavailable dataset exists merely because such a dataset would be scientifically desirable. The empirical question is instead:

$$\text{What data exist} \longrightarrow \text{what data are observable} \longrightarrow \text{what can be independently validated.} \quad (55)$$

## 22 The India–China Application

The India–China border provides a natural case study because it contains multiple dated episodes of confrontation, standoff, disengagement, and subsequent strategic interaction.

The 2017 Doklam episode is particularly useful as a historical observation because it supplies a dated confrontation that did not become a general war. Such observations are indispensable for avoiding a selection problem in which only escalatory cases are studied.

The empirical sample should therefore contain both

$$\text{confrontation} \longrightarrow \text{escalation} \quad (56)$$

and

$$\text{confrontation} \longrightarrow \text{de-escalation.} \quad (57)$$

The objective is not to force historical events into the theory. It is to determine whether the theory survives contact with historical data.

## 23 Limitations

Several limitations are fundamental.

First, the Ghosh Factor application is presently cross-sectional and therefore cannot by itself establish temporal causality.

Second,  $Z$  and  $Y$  contain constructed weights. These weights require calibration or estimation rather than arbitrary selection.

Third, military data can be incomplete, classified, strategically manipulated, or measured differently across states.

Fourth, skirmish observations are subject to reporting bias.

Fifth, geopolitical influence is partly relational and therefore cannot always be reduced to a state-level scalar.

Sixth, machine-learning models can discover predictive associations without identifying causal mechanisms.

Seventh, conflict is a low-frequency event. Consequently, apparently high predictive accuracy can be misleading when the base rate of escalation is low.

These limitations imply that the framework should be treated as an empirical research programme rather than a completed theory of war.

## 24 Conclusion

This paper has integrated unequal economic growth, geopolitical geometry, militaric capability, and skirmish dynamics into one framework.

The central chain is

$$\boxed{\text{Unequal Growth} \rightarrow \text{Economic Capacity} \rightarrow \{Z, Y\} \rightarrow \Delta \rightarrow \text{Skirmish} \rightarrow \text{Conflict Hazard}}. \quad (58)$$

The principal theoretical object is the geopolitical–militaric gap

$$\Delta_i = |Z_i - Y_i|. \quad (59)$$

The principal empirical objective is to determine whether this gap, together with economic growth, geographic imbalance, diplomatic relationships, and historical skirmish observations, contains statistically useful information about subsequent conflict dynamics.

The most important methodological principle is temporal discipline. A model that explains historical observations in-sample is not necessarily a model that predicts future observations. The decisive experiment therefore requires longitudinal data, temporal train–validation–test separation, transparent measurement, uncertainty quantification, and comparison against simple baselines.

The framework ultimately connects economics and international relations through a common resource constraint: economic development determines part of the resource envelope from which geopolitical projection and militaric capacity are constructed. Conflict emerges when political objectives, geopolitical perceptions, military capacity, economic resources, and information interact under uncertainty.

The empirical question is consequently not whether history can be rewritten. It is whether history can be measured well enough that its structure becomes useful information for understanding the future.

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# Glossary

## **Ghosh Factor**

A conditional logarithmic transformation used to model deviations around a reference growth state.

## **Geopolitical Score $Z$**

A composite measure based on military size, economic output, and population, with directional weighting in the geometric formulation.

## **Militaric Score $Y$**

A composite measure incorporating military size, economically available wealth, and the civilian demographic base.

## **Skirmish Gap**

The absolute divergence  $\Delta = |Z - Y|$  between geopolitical influence and militaric capability.

## **Operational Largeness**

A coefficient representing the scale at which latent geopolitical–militaric pressure becomes observable as skirmish activity.

## **Attenuation Rate**

The rate at which skirmish intensity decreases through time in the absence of new pressure.

## **Angular Imbalance**

The directional separation between geopolitical relationships represented on the geopolitical circle or sphere.

## **Conflict Hazard**

The conditional probability rate of a specified future conflict event.

## **State Capacity**

The economically, demographically, institutionally, and militarily available resource base of a state.

## **Temporal Validation**

An evaluation procedure in which observations from the future are withheld from model estimation and used only for subsequent testing.

## **Look-ahead Bias**

A statistical error caused by allowing information that would not have been available at the prediction date to influence model estimation.

## **Graph Learning**

Machine learning in which states are represented as nodes and relationships among states as edges.

## **Political Risk**

The probability-weighted economic consequence of political or geopolitical events.

## **Warfare Economics**

The study of resource allocation, incentives, production, welfare losses, and economic constraints associated with military activity and war.

# The End